

Ordinary Short-Seed Collisions in the Unsalted TRINE PDF Construction: Equivalent Signing-Witness Recovery and Fresh-Message Forgery

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Provenance and review status. The TRINE-specific attack instantiation, derivation, implementation, experiments, resource analysis, and initial manuscript were produced by OpenAI Codex using Daybreak Blue at Ultra reasoning effort during an AI-run audit of NGCC signature candidates. A separately instructed hostile-review agent reconstructed the attack, corrected the resource accounting, rebuilt the full-dimension reduced-seed experiments, and tried to disprove the result. At the date above, no human cryptanalyst has independently verified the argument, implementation, measurements, or novelty assessment. This report is a preliminary disclosure for human review and reproduction.

Abstract

The submitted TRINE PDF specification compresses every base opening in a signature to a λ -bit seed. Algorithms 6–8 and the signature-size formula contain no per-signature salt, and the point sampled from that seed is fixed before the Fiat–Shamir hash sees the message. Under the direct, message-independent realization of this PDF construction, ordinary independently randomized signatures eventually reuse a round seed at the same round index. If the two Fiat–Shamir labels differ, the public canonical forms are equal. The public canonicalization certificates then give a complete equivalence between the two corresponding trilinear forms.

For level-I Balanced, one base/nonbase collision recovers an equivalent signing witness. For level-I ShortSig, collision edges between the five form labels compose; any connected label graph recovers all four maps from the base form. The recovered first-coordinate maps suffice to answer every future challenge, so the attacker signs arbitrary fresh messages without the original secret key. Full-dimension experiments reduced only the round-seed entropy to make natural birthday collisions executable. They used fresh independent DRBG output, recovered and checked the complete public tensor equivalences, erased the original secret material, and produced fresh-message signatures accepted by a PDF-conformant verifier. Wrong-message, response-tamper, digest-tamper, and corrupted-map controls rejected.

Most directly, set the signing-query count to $Q = 2^{64} - 1 < 2^{64}$. In the exact fixed-index ideal model, the least public tables giving success at least one half contain $2^{64.879318}$ and $2^{67.639106}$ distinct seeds, with one 256-bit canonical fingerprint per seed, for Balanced-I and ShortSig-I. The corresponding returned traffic is $2^{78.609179}$ and $2^{77.661778}$ bits. A raw 48-byte seed/fingerprint table occupies $2^{70.464281}$ and $2^{73.224068}$ bytes; conservative 80-byte records occupy $2^{71.201246}$ and $2^{73.961034}$ bytes. Measured full-canonical-form accounting gives $2^{94.246627}$ and $2^{96.990188}$ attacker cycles, while the honest signing service performs $2^{97.268224}$ and $2^{96.182868}$ cycles. Thus the attack algorithms have full-parameter cost estimates below the claimed 128-bit level and need fewer than 2^{64} signing queries, but they are not physical

2^{80} attacks. Only the reduced-seed end-to-end forgeries were executed; no full-entropy forgery was run.

The untouched submitted C source serializes a fresh 2λ -bit salt and absorbs it in every round expansion. That source-only field is absent from the PDF algorithms and size formula, makes level-I signatures 32 bytes longer, and blocks this cross-signature accumulation. We therefore claim an attack only on the direct unsalted, message-independent PDF construction, not on the untouched salted source. The generic short-seed collision attack and the salt-plus-round repair are due to Beullens and Sipasseuth. Our contribution is limited to the TRINE instantiation: the PDF/source discrepancy, canonical- certificate extraction, one-edge Balanced completion, connected-graph ShortSig completion, exact level-I resource analysis, and end-to-end validation.

1 Introduction

TRINE is a Fiat–Shamir signature based on trilinear-form equivalence (TFE) [1]. Its identification protocol commits to canonical forms computed from effective corank-one points. A base challenge opens the point itself; a nonbase challenge opens the point transported through a secret equivalence. The canonical form is both an invariant and an extractor: two different openings of one commitment reveal an equivalence between the public forms.

The PDF reduces signature size by replacing each base opening with a λ -bit seed. This creates a finite commitment-randomness domain. With many ordinary signatures, a seed repeats. When the Fiat–Shamir labels at that round differ, the special-soundness event needed to recover a public-form equivalence occurs naturally. Balanced needs one such event. ShortSig has four nonbase forms, but its recovery problem is only connectivity of a five-vertex graph: edges from different round indices compose because the secret equivalences are profile-wide.

Short-seed collision attacks of this kind are known. Beullens described a public-table attack on ALTEQ and the need to include a salt and round identifier in the expansion domain [2]. Sipasseuth’s DVA analysis explicitly discusses both attacker-generated seed tables and internal collisions, together with the same repair [3, Sec. 3.2]. This report does not claim those generic ideas. It establishes that the unsalted algorithm and size formula in the TRINE PDF instantiate the attack, derives the exact extraction and graph completion for both level-I profiles, and accounts for the unusually expensive TRINE seed decoder.

Our contributions are as follows.

1. We give the exact same-index collision and canonical-certificate extraction equations for the PDF construction.
2. We prove that one collision edge is a complete Balanced signing witness and that any connected collision graph is a complete ShortSig signing witness.
3. We give an exact fixed-index ideal-model attack point at $Q = 2^{64} - 1$ and, separately, derive approximate all-round sparse-Poisson sensitivity estimates and broader time–memory–data tradeoffs, while accounting separately for signing service, seed decoding, canonicalization, returned traffic, retained storage, and peak RAM.
4. We implement the full extraction-to-forgery chain at the exact level-I dimensions, reducing only seed entropy. A second implementation based on the submitted source uses ordinary fresh DRBG output and exact PDF signature sizes.
5. We isolate the source/PDF discrepancy. The untouched source already contains the fresh 2λ -bit salt needed to defeat accumulation; it is outside the attack claim and is not vulnerable to this mechanism.

The attack is unconditional with respect to signer state: it uses neither rollback nor repeated-state control, faults, related keys, rare keys, nor a chosen-randomness interface. Its scope does

depend on the PDF seed decoder being deterministic and message-independent. That is the direct reading of the written sampling order and matches the submitted decoder after removing only its source-only salt. A hypothetical decoder keyed by the message would define a different construction and defeat cross-message accumulation.

2 The relevant TRINE construction

2.1 Public forms and the group action

Let $\phi : \mathbb{F}_{q_{\text{fld}}}^n \times \mathbb{F}_{q_{\text{fld}}}^n \times \mathbb{F}_{q_{\text{fld}}}^n \rightarrow \mathbb{F}_{q_{\text{fld}}}$ be a trilinear form. For $T = (A, B, C) \in \text{GL}(n, q_{\text{fld}})^3$, write

$$\phi^T(x, y, z) = \phi(Ax, By, Cz). \quad (1)$$

The action is on the right:

$$(\phi^T)^S = \phi^{TS}. \quad (2)$$

TRINE publishes $X + 1$ equivalent forms. We use label X for the base form:

$$\phi_j = \phi_X^{E_j}, \quad E_j = (A_j, B_j, C_j), \quad 0 \leq j < X, \quad qqquad E_X = (I, I, I). \quad (3)$$

The original secret key contains the nonbase equivalences E_j . Any other triples satisfying (3) are equivalent signing witnesses; recovering the original key representation is unnecessary.

For an effective corank-one point a of a form ϕ , TRINE computes the canonical form $\text{CF}(\phi, a)$. The public canonicalization procedure also constructs invertible bases. We denote its certificate by $T_{\phi, a} = (P, Q, R)$, so that

$$\phi^{T_{\phi, a}} = \text{CF}(\phi, a). \quad (4)$$

2.2 Signing and verification equations

Algorithm 6 first samples all round points and commitments, and only then hashes the message:

$$\begin{aligned} a_i &\leftarrow \$ \text{Corank1Cal}(\phi_X), \\ \psi_i &= \text{CF}(\phi_X, a_i), \end{aligned} \quad 0 \leq i < r, \quad (5)$$

$$\begin{aligned} cha &= H(M \parallel \psi_0 \parallel \dots \parallel \psi_{r-1}), \\ (b_0, \dots, b_{r-1}) &= \text{ParseHash}(cha), \\ d_i &= A_{b_i}^{-1} a_i. \end{aligned} \quad (6)$$

The parser chooses exactly K nonbase positions. Conditional on a nonbase position, its label is uniform in $\{0, \dots, X - 1\}$; all remaining positions have label X .

Verification reconstructs $\text{CF}(\phi_{b_i}, d_i)$, rehashes those forms with the message, reparses the challenge, and accepts only if the challenge and every opening are consistent. Correctness follows from

$$\text{CF}(\phi_j, A_j^{-1} a) = \text{CF}(\phi_X, a). \quad (7)$$

Section 3.4 permits a base point a_i to be sent as a seed. Section 3.5 assigns exactly λ bits to each of the $r - K$ base seeds and gives

$$|\sigma| = (r - K + 2)\lambda + Kn \lceil \log_2 q_{\text{fld}} \rceil. \quad (8)$$

There is no salt in Algorithms 6–8 or (8). A conservative direct realization preserves round separation through public maps

$$D_i : \{0, 1\}^\lambda \longrightarrow \{\text{effective corank-one points of } \phi_X\}, \quad \psi_i(s) = \text{CF}(\phi_X, D_i(s)). \quad (9)$$

The attack compares seeds only at equal indices i ; it never assumes a common decoder across round indices. Injectivity is unnecessary: equal inputs always give equal commitments, while additional decoder collisions only help.

Normative decoder evidence. The direct message-independent reading of D_i rests on cumulative evidence in the submission. Algorithm 6 samples $a_i \leftarrow \$ \text{Corank1Cal}(\phi_X)$ and computes every ψ_i before hashing the message, and the point sampler has no message argument. Section 3.4 permits a base opening to be compressed to a seed; Section 3.5 assigns exactly λ bits to each such seed. Algorithms 6–8 and the PDF signature-size formula serialize no salt. In contrast, the submitted C source adds a fresh 2λ -bit salt, absorbs it in every round expansion, and consequently produces level-I signatures 32 bytes longer.

2.3 Level-I parameters

Table 1 gives the two targeted profiles. The PDF sizes follow directly from (8). The submitted source sizes include one extra 2λ -bit field.

Table 1: TRINE level-I parameters and signature sizes.

profile	n	q_{fld}	r	K	X	PDF/source bytes
Balanced-I	22	4093	138	52	1	3,124 / 3,156
ShortSig-I	22	4093	61	36	4	1,620 / 1,652

An exhaustive search of the complete 23-page specification, its EUF-CMA and parameter-selection sections, the basic-information form, both IP documents, the implementation README and interface notes, all reference and optimized source trees, and the KAT material found no maximum signatures per key and no mandatory key-rotation rule. In particular, KeyGen has no lifetime input, Section 5.2 states EUF-CMA security without a query cap, and Section 6.3 assigns the 128-bit level from (r, K, X) without a lifetime condition. This establishes only the absence of a specified cap; it does not attribute an affirmative unlimited-lifetime policy to the submitters. The $Q = 2^{64} - 1$ point below is therefore not excluded by a submitted per-key lifetime rule. A deployment enforcing a lower cap cannot reach that particular point. Its other resources remain vastly beyond physical execution on the audit machine.

3 Collision extraction

Lemma 1 (Same-index seed collision). *Suppose two signatures use the same underlying round seed s at one round index i , but their challenge labels there are c and d . Under (9), their verifier-reconstructed canonical forms are equal:*

$$\text{CF}(\phi_c, u_c) = \psi_i(s) = \text{CF}(\phi_d, u_d).$$

This holds for ordinary independent signing randomness; it does not require the signer state to repeat.

Proof. Both signing calls evaluate the same deterministic map D_i on s , giving the same base point and commitment. Correctness (7) carries that commitment through either challenge response. Repetition arises from the finite λ -bit seed space under independent sampling. $\square$

Lemma 2 (Public certificate extraction). *Let $c \neq d$ and suppose public openings u_c, u_d satisfy $\text{CF}(\phi_c, u_c) = \text{CF}(\phi_d, u_d)$. Let T_c, T_d be the public certificates from (4). Then*

$$\phi_c^{T_c T_d^{-1}} = \phi_d. \tag{10}$$

Proof. The equal canonical form gives $\phi_c^{T_c} = \phi_d^{T_d}$. Applying T_d^{-1} on the right and using (2) proves the claim. $\square$

The attacker detects a candidate edge by comparing canonical-form encodings, not by observing hidden seed values. Before accepting an edge, the reproducer byte-compares the complete canonical tensors and checks (10) against all public tensor coefficients. Stabilizers or nonunique certificates do not matter: any extracted triple that passes this public equation is a valid equivalence.

3.1 Balanced: one edge

Balanced has labels $\{0, 1\}$ with base label $X = 1$. A collision with $c = 1, d = 0$ yields

$$E'_0 = T_1 T_0^{-1}, \quad \phi_1^{E'_0} = \phi_0.$$

Let A'_0 be its first component. On every future nonbase challenge the attacker returns $(A'_0)^{-1}a_i$; on a base challenge it serializes the seed.

Theorem 3 (Balanced completion). *One observable same-index collision under the two different Balanced labels gives an equivalent signing witness and permits valid signatures on arbitrary fresh messages.*

Proof. Lemma 2 produces a publicly checked triple mapping the base form to the only nonbase form. Equation (7) shows that its inverse first component answers every nonbase response. All other signing steps use public algorithms and fresh attacker-chosen round seeds. The Fiat–Shamir hash therefore binds a correctly opened commitment vector to the new message. $\square$

3.2 ShortSig: connected-graph completion

ShortSig has labels $V = \{0, 1, 2, 3, 4\}$ with base label 4. Each collision between distinct labels c, d yields the directed equivalence $E_{c \rightarrow d} = T_c T_d^{-1}$ and its inverse. For a path

$$4 = v_0, v_1, \dots, v_m = j,$$

right-action composition gives

$$E'_{4 \rightarrow j} = E_{v_0 \rightarrow v_1} E_{v_1 \rightarrow v_2} \cdots E_{v_{m-1} \rightarrow v_m}, \quad \phi_4^{E'_{4 \rightarrow j}} = \phi_j. \quad (11)$$

Theorem 4 (ShortSig completion). *If the graph of observed cross-label seed collisions is connected, the attacker recovers an equivalent signing witness for all four ShortSig nonbase forms and can sign arbitrary fresh messages.*

Proof. Connectivity supplies a path from base vertex 4 to every nonbase vertex. Equation (11) produces and publicly verifies every required equivalence. Inverting the first component of each triple supplies the response map in (6). The rest follows as in Theorem 3. $\square$

Edges may come from different round indices. Round separation changes which seed values collide, but it does not change the profile-wide public-form maps recovered from those edges.

4 Collision acquisition and success probabilities

4.1 All-round internal collisions

Process ordinary signatures once, reconstruct every round canonical form, and compare records only within the same index. Let

$$p_j = \frac{K}{rX} \quad (j < X), \quad p_X = \frac{r - K}{r}. \quad (12)$$

For Q signatures, the sparse-Poisson edge intensity between labels u, v is

$$\mu_{uv} = \frac{Q(Q-1)rp_up_v}{2^{128}}. \quad (13)$$

For Balanced, success means at least one edge, so

$$\Pr[\text{success}] \approx 1 - \exp\left(-\frac{Q(Q-1)rp_0p_1}{2^{128}}\right). \quad (14)$$

Its 50-percent point is $\log_2 Q = 61.226527$.

For ShortSig, assign independent edge probabilities $1 - e^{-\mu_{uv}}$ in the sparse approximation and sum the probabilities of the connected graphs among the $2^{\binom{5}{2}} = 2^{10}$ edge sets. The resulting median is $\log_2 Q = 62.990592$. Nonbase/nonbase edges materially improve on a base-star-only collection.

These all-round probabilities are an ideal sparse-Poisson model, not exact probabilities for conditioned serialized signatures. The labels within one signature have fixed weight and are correlated. An exact fixed-weight toy simulation with 16-bit seeds obtained 968 successes in 2,000 Balanced trials at 37 signatures, versus model probability 0.49183, and 973 successes in 2,000 ShortSig trials at 127 signatures, versus 0.49811. Digest collisions and random-oracle conditioning are a separate negligible layer in this model.

4.2 Fixed-index public table

Fix one round i . Decode and canonicalize P distinct public seeds under the base form, then inspect the opening at index i in each of Q ordinary signatures. A nonbase opening whose canonical form appears in the table gives an edge.

Let $N = 2^{128}$ and $p = K/r$. For Balanced, the exact uniform-seed, ideal-challenge probability is

$$1 - \left(1 - \frac{pP}{N}\right)^Q. \quad (15)$$

For ShortSig, requiring a hit for all four nonbase labels gives by inclusion–exclusion

$$\sum_{s=0}^4 (-1)^s \binom{4}{s} \left(1 - \frac{spP}{4N}\right)^Q. \quad (16)$$

These are exact for uniform seeds and independent ideal labels at the selected index; they remain analytic models for the concrete Fiat–Shamir transcript.

At sparse 50-percent points,

$$QP = \frac{zXN}{p}, \quad z = \ln 2 \text{ for Balanced}, \quad z = -\ln(1 - 2^{-1/4}) \text{ for ShortSig}. \quad (17)$$

This exposes the full time–memory–data tradeoff. If decoding one public seed costs c_D and processing one nonbase target costs c_F , measured attacker work is

$$W(P) = c_DP + pQc_F. \quad (18)$$

Proposition 5 (A fixed-index point below 2^{64} signing queries). *Set*

$$Q = 2^{64} - 1 = 18,446,744,073,709,551,615.$$

For Balanced-I and ShortSig-I, respectively, set

$$\begin{aligned} P_B &= 33,932,896,019,960,893,868, \\ P_S &= 229,826,321,259,624,867,428. \end{aligned}$$

Substitution in the exact ideal-model formulae (15) and (16) gives, respectively,

$$\begin{aligned}\Pr_B[\text{success}] &= 0.5000000000000000000044518893\dots, \\ \Pr_S[\text{success}] &= 0.5000000000000000000011550258\dots\end{aligned}$$

Thus both specified witness-recovery routes have approximately 50-percent success with strictly fewer than 2^{64} signing queries.

Proof. Use $p = 52/138$ in (15), and $p = 36/61$ in (16). The companion script evaluates those formulae with 120 decimal digits and integer binary search. It additionally obtains $0.4999999999999999999942383907\dots$ at $P_B - 1$ and $0.4999999999999999999981283883\dots$ at $P_S - 1$. Hence the displayed integers are the least table sizes reaching one half under the stated exact fixed-index ideal model. No sparse-Poisson approximation enters this result. For ShortSig the four hits make a base-centered star, which is connected, so Theorem 4 applies. $\square$

This point matters independently of its large physical work: it does not require 2^{64} signing queries, and the submission specifies no lower per-key lifetime cap. It uses ordinary independently randomized signatures.

5 Concrete resources

Every exponent in this section is base two. The figures use the level-I PDF signature sizes. “Signer cycles” charge all signing-oracle service separately using the cycle counts in the submitted performance table. “Attacker cycles” use native measurements on an AMD EPYC host at 1.996249 GHz. Parallelism changes wall time, not aggregate work.

5.1 Measured primitive costs

The decoder repeatedly samples field points until `Corank1Cal` and canonicalization succeed. A success occurs only about once per 4,093 candidates. The hostile benchmark measured the complete seed-to-canonical-form path:

operation	Balanced-I	ShortSig-I
seed $\rightarrow$ <code>Corank1Cal</code> /retry $\rightarrow$ CF, seconds	0.345799062	0.342778750
seed $\rightarrow$ <code>Corank1Cal</code> /retry $\rightarrow$ CF, cycles	$2^{29.36265}$	$2^{29.34999}$
full CF of an exposed point	0.005459163 s ($2^{23.37754}$ cycles)	
12-value partial invariant	0.000599711 s ($2^{20.19120}$ cycles)	

The seed measurement already includes the full canonical form. Omitting this retry path understates the attack by many bits.

5.2 The $Q = 2^{64} - 1$ security point

Table 2 accounts for Proposition 5. Each public-table entry contains one distinct 16-byte seed and one 32-byte canonical-form fingerprint. The raw table therefore uses 48 bytes per entry. The second storage row conservatively allows 80 bytes for sorting and retrieval metadata, as in the all-round accounting. “Query-side hashes” is the expected number pQ of nonbase openings at the selected index that must be canonicalized and fingerprinted. On a fingerprint match, the attacker regenerates and byte-compares both complete 15,972-byte canonical tensors and accepts an edge only after checking the public tensor-equivalence equation coefficient by coefficient.

The full-canonical attacker count is

$$P(c_D \cdot 1.996249 \cdot 10^9) + pQ(c_F \cdot 1.996249 \cdot 10^9)$$

Table 2: Exact fixed-index ideal-model points with fewer than 2^{64} signing queries. Resource entries other than Q and success are base-two exponents.

resource	Balanced-I	ShortSig-I
signing queries Q	$2^{64} - 1$	$2^{64} - 1$
public table seeds	$2^{64.879318}$	$2^{67.639106}$
public table hashes	$2^{64.879318}$	$2^{67.639106}$
expected query-side hashes	$2^{62.591915}$	$2^{63.239188}$
returned signature bits	$2^{78.609179}$	$2^{77.661778}$
raw table bytes (48/entry)	$2^{70.464281}$	$2^{73.224068}$
operational table bytes (80/entry)	$2^{71.201246}$	$2^{73.961034}$
measured attacker cycles	$2^{94.246627}$	$2^{96.990188}$
honest signer cycles	$2^{97.268224}$	$2^{96.182868}$
exact ideal success	$0.5 + 4.45 \cdot 10^{-21}$	$0.5 + 1.16 \cdot 10^{-21}$

cycles, with c_D and c_F from the measured table above. The signer count is Q times the submission’s reported signing cycles. These are aggregate work figures; parallel execution changes wall time but not the displayed cycle totals. The exact probability belongs to the uniform-seed, ideal-challenge model at one fixed index. It is not asserted as the exact probability of a conditioned serialized Fiat–Shamir transcript.

5.3 All-round route

Each signature yields r externally sortable records. The attacker decodes every base seed and canonicalizes each exposed nonbase point. Table 3 reports the median point.

Table 3: All-round internal-collision resources.

resource	Balanced-I	ShortSig-I
signing queries	$2^{61.226527}$	$2^{62.990592}$
returned signature bits	$2^{75.835706}$	$2^{76.652370}$
round records / canonical results	$2^{68.335051}$	$2^{68.921330}$
retained bytes at 80 bytes/record	$2^{74.656979}$	$2^{75.243258}$
measured attacker cycles	$2^{97.029148}$	$2^{97.017155}$
honest signer cycles	$2^{94.494751}$	$2^{95.173460}$

The 80-byte record contains a canonical digest, label, round index, opening or seed, and retrieval metadata. External sorting permits an operator-selected RAM chunk; peak RAM need not approach the retained-disk exponent. On a digest match, the attacker regenerates and byte-compares both 15,972-byte canonical forms before extraction. A record-hash collision can add work but cannot produce a false witness.

5.4 Fixed-index Pareto points

Table 4 separates three useful points. The first minimizes the number of fingerprint evaluations and keeps raw traffic just below 2^{80} bits. The second minimizes measured aggregate attacker cycles with full canonical forms and is the rigorous CPU baseline. The third holds signing queries at the all-round medians, minimizing traffic but requiring a much larger public table.

At the CPU minimum, the public seed table is much smaller but the oracle returns more data and performs roughly $2^{101.4}$ cycles of signing work. At the data-oriented point, attacker work rises to $2^{93.12}$ and $2^{94.81}$ cycles. No row is a simultaneous 2^{80} time/data/memory attack.

Table 4: Fixed-index full-canonical-form Pareto points. Columns headed B and S are Balanced-I and ShortSig-I.

resource	minimum calls		minimum attacker CPU		all-round Q	
	B	S	B	S	B	S
$\log_2 Q$ queries	65.1437	66.2000	68.1363	69.1862	61.2265	62.9906
$\log_2 P$ table seeds	63.7356	65.4391	60.7430	62.4529	67.6528	68.6485
returned bits	79.7529	79.8618	82.7455	82.8480	75.8357	76.6524
table bytes	70.0575	71.7610	67.0649	68.7748	73.9747	74.9704
attacker cycles	93.1209	94.8119	91.1057	92.8029	97.0155	97.9988
signer cycles	98.4119	98.3829	101.4045	101.3691	94.4948	95.1735

5.5 Partial invariant acceleration

The canonicalizer’s alternating kernel chain produces vectors that transform up to independent nonzero scales. Let $T_{ijk} = \phi(u_i, v_j, w_k)$ and $d = T_{010}$. When the denominators are nonzero, the ratios

$$\begin{aligned} \frac{dT_{ij0}}{T_{i10}T_{0j0}} & \quad (i, j \in \{2, 3\}), \\ \frac{d^2T_{ijk}}{T_{i10}T_{0j0}T_{01k}} & \quad (i, j, k \in \{2, 3\}) \end{aligned} \quad (19)$$

cancel every chain-vector scale. The resulting 12 elements of $\mathbb{F}_{4093}$ have a nominal 143.987 bits of output.

An independent full-dimension primitive test obtained equality for 64/64 planted equivalences and 0/64 independent controls, with a 9.10-fold speedup over full canonicalization. Equivariance is exact, but the tuple’s generic discrimination entropy on TRINE’s structured distribution is not proved. Every hit must therefore receive full-CF confirmation. Treating the tuple as a high-entropy prefilter gives $2^{89.51}$ and $2^{91.21}$ measured cycles at its CPU-optimized points; these are heuristic sensitivity figures. The $2^{91.11}$ and $2^{92.80}$ full-CF rows remain the rigorous fixed-index baselines.

6 End-to-end validation

Full-entropy PDF conformance. A clean copy of the submitted reference source received only the minimal salt-removal and signature-layout patch described below. The round seed remained 16 bytes (128 bits); no collision instrumentation or entropy reduction was compiled. One ordinary Balanced-I signature verified and opened at exactly 3,124 bytes, and one ordinary ShortSig-I signature verified and opened at exactly 1,620 bytes. Changed-message controls rejected. This is a full-entropy conformance test of the direct unsalted PDF transcript, not a collision, extraction, or forgery experiment. It establishes that reducing seed entropy is needed only to make the birthday events experimentally reachable.

Reduced-seed attack validation. Two implementations tested the complete extraction-to-forgery chain while retaining the exact level-I dimensions $(n, q_{\text{fld}}, r, K, X)$ and reducing only seed entropy.

Portable tests. With eight-bit independent fresh seeds, Balanced needed four ordinary signatures and ShortSig needed nine in the frozen runs. Both tests detected collisions through public canonical equality, recovered the required tensor maps, erased the planted signing maps,

and forged a fresh message using only the recovered witness. Wrong-message and corrupted-map controls rejected.

Submitted-source hostile harness. A separately written harness aligns the submitted level-I core with the PDF transcript, retains round-index separation, and reduces the seed domain to six bits. For instrumentation convenience, the former salt input is fixed to the public all-zero string and omitted from serialization; a fixed public value adds no per-signature decoder entropy. The bundle separately includes the minimal conformance patch that removes the source-only salt from the decoder as well. The submitted DRBG is explicitly initialized and never reset or rolled back. Across two independently generated keys per profile, Balanced needed two signatures for each key; ShortSig needed five and six. All four fresh-message forgeries were accepted at the exact PDF sizes 3,124 and 1,620 bytes after the original secret and collected transcript state were erased. Changed-message, response-tamper, and digest-tamper controls all rejected.

For every extracted edge, both implementations compare the complete canonical tensors and verify the corresponding public tensor-equivalence equation, not merely a canonical digest. The Balanced run then checks its single recovered base-to-nonbase map. The ShortSig run checks every edge and every composed base-to-label map before using connected-graph completion. These runs establish collision observability, certificate orientation, graph composition, witness sufficiency, verifier binding, and ordinary fresh randomness. They do not estimate or execute the full 128-bit-domain attack; those costs come from the separately qualified analytic models.

7 Specification and implementation boundary

The untouched submitted source expands each round from

$$\text{"MEDS2END-TRINE-ROUND-v1"} \parallel salt \parallel s_i \parallel \text{LE32}(i), \quad (20)$$

where *salt* is a fresh serialized 2λ -bit value. Its level-I signatures are therefore 32 bytes longer than the PDF values in Table 1. Independent salts make the decoder domain signature-specific, so equality of the serialized round seed no longer gives equality of the expanded point. The ordinary attack in this report does not apply to that source construction: the untouched submitted C source is not vulnerable to this accumulation mechanism. This report is not an implementation-break claim. It targets the direct unsalted, message-independent realization written in the PDF, while the source already contains the relevant salt-based repair.

The reproducer contains two patches. The minimal conformance patch removes only the source-only salt from sampling, serialization, codecs, and the round XOF while preserving the domain string, round index, SHAKE256, field sampler, Corank1Cal retry logic, canonicalizer, challenge parser, and verifier. A second patch adds only reduced-seed and certificate instrumentation for the hostile experiment.

The PDF leaves the precise seed decoder implicit. A contrary interpretation could define $D_i(s, M)$, binding the message without adding signature bytes. That construction would defeat cross-message accumulation. It is not the written order of Algorithm 6, where $a_i \leftarrow \$\text{Corank1Cal}(\phi_X)$ precedes the message hash and has no message argument, and it is not the submitted decoder. For this reason, the exact claim is an ordinary collision attack on the *direct unsalted, message-independent PDF realization*. If the submitted salt is declared normative, or a message-bound decoder is specified, the claim does not apply. Nothing in the analysis treats a source deviation as a vulnerability; the minimal patch is used only to test the PDF construction that the paper analyzes.

8 Repair

The PDF should specify a per-signature public salt of at least 2λ bits, serialize it, include it in the signature-size formula, and bind it together with the round index and full instance identifier before any round expansion. For example,

$$D_{inst,i,\rho}(s) = \text{Decode}(\text{XOF}(\text{domain} \parallel inst \parallel \rho \parallel s \parallel \text{LE32}(i))). \quad (21)$$

The source already implements the central salt-and-round part of this repair. Alternatively, binding the complete message digest into the decoder prevents cross-message accumulation, but the decoder and proof must then state that dependency explicitly. Tests should assert that equal serialized round seeds under two independently salted signatures produce unrelated commitments.

9 Prior art and novelty boundary

Ward Beullens’s public note *Trivial multi-key attacks + attack on ALTEQ*, posted to the NIST PQC Forum on 21 February 2024, describes the generic public-table collision/guessing attack on λ -bit signing seeds and recommends deriving rounds from the seed together with a salt and round identifier [2]. Arnaud Sipasseuth’s *DVA: Dangerous Variations of ALTEQ*, IACR ePrint 2024/817, Section 3.2, explicitly includes internal seed collisions and the *salt* $\parallel$ *rid* repair [3]. They own the generic mechanism and repair.

The contribution claimed here is narrower: applying that mechanism to the unsalted TRINE PDF; identifying the discrepancy with the already salted source; using TRINE’s public canonicalization certificate to extract a full tensor equivalence; showing that Balanced needs one edge while ShortSig needs a connected five-label graph; deriving the exact level-I tradeoffs with mandatory *Corank1Cal* retry costs; and validating fresh forgeries at full TRINE dimensions with reduced seed entropy. The connected-graph repetition is a candidate-specific completion of the known per-label attack, not a new generic attack family.

A best-effort check of the public NGCC TRINE page on 3 October 2026 found only report **sign-30-1**, an unrelated deterministic-key implementation failure [4]. No public TRINE-specific report of the unsalted-PDF seed-domain attack was found at that checkpoint. This is a dated best-effort negative search, not proof of novelty or absence.

10 Reproduction

The companion ZIP is self-contained and contains the PDF-aligned source, minimal conformance patch, hostile instrumentation patch, Balanced and ShortSig attack harnesses, portable controls, resource model, frozen outputs, validator, and checksum manifest. From its **reproducer** directory, run

```
./run_all.sh.
```

The two full-dimension reduced-seed profile runs execute concurrently. On the audit host, the slower frozen run took approximately eight minutes and used about 72 MiB peak resident memory. The command ends with **validation=PASS**. To check the sealed evidence and manifest without rerunning the native experiments, use `./run_all.sh -validate-only`. No command depends on an audit-cache or private workspace path.

11 Conclusion

The direct unsalted TRINE PDF construction places every base commitment in a 2^{128} seed domain. Ordinary independent signatures eventually collide at one round index. Different

challenge labels turn that birthday event into the canonical-form equality that TRINE’s own public certificates extract. One edge completes Balanced; a connected label graph completes ShortSig. The resulting equivalent witness creates ordinary fresh-message signatures.

The attack algorithms cover the complete extraction-to-forgery chain and give full-parameter cost estimates, but a full-entropy forgery was not physically executed. In particular, the exact fixed-index point at $Q = 2^{64} - 1$ has approximately one-half ideal-model success and is not excluded by any specified TRINE per-key lifetime cap. Its measured attacker work is about $2^{94.25}$ and $2^{96.99}$ cycles, and signer work, traffic, and storage are reported separately. Other Pareto points reduce attacker work or query data at different physical costs. Only reduced-seed end-to-end forgeries were run. The untouched source is protected by a fresh serialized 2λ -bit salt, is not vulnerable to this accumulation mechanism, and is outside the claim. The generic attack belongs to Beullens and Sipasseuth; the result here is the concrete unsalted-PDF TRINE instantiation and its validated extraction-to-forgery chain.

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References

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